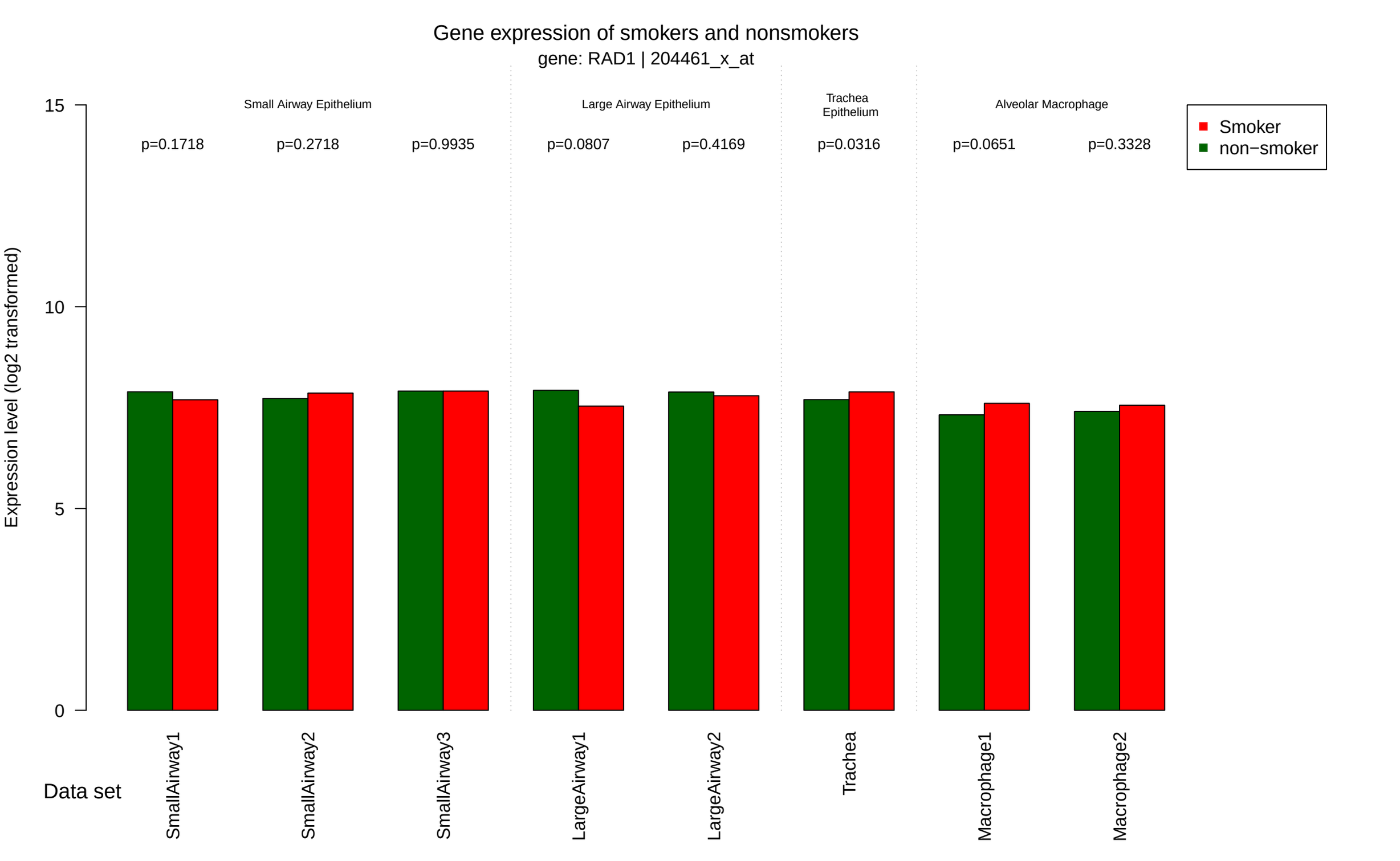
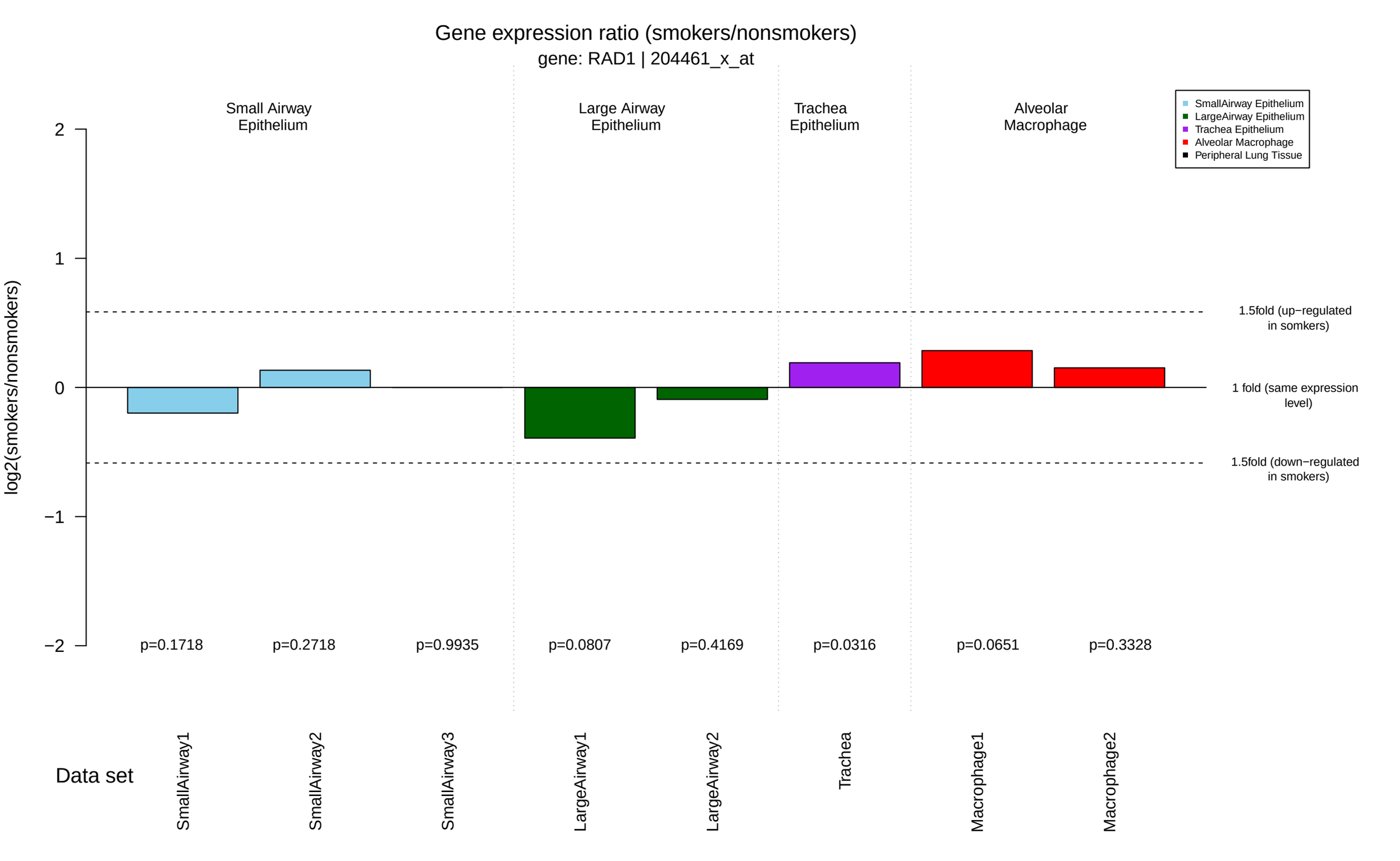
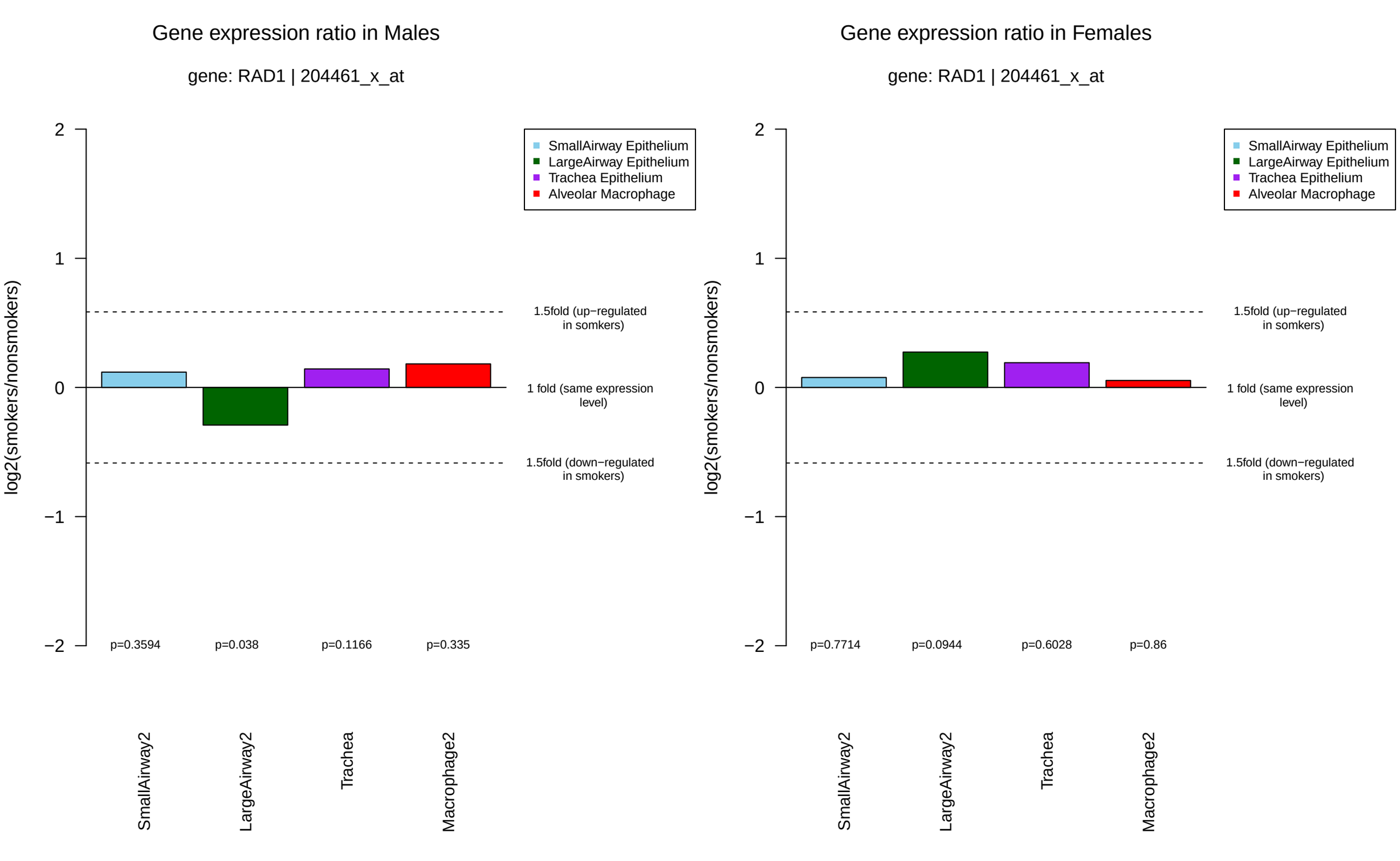
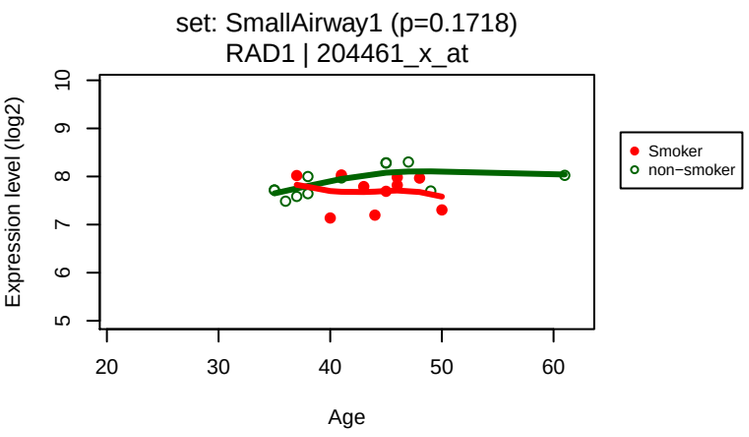
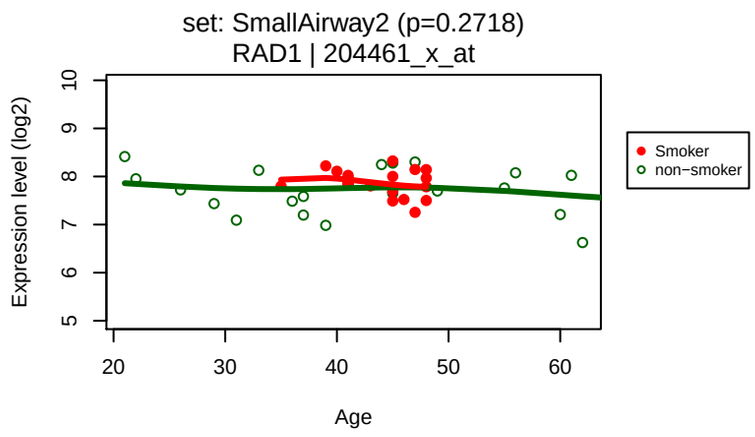
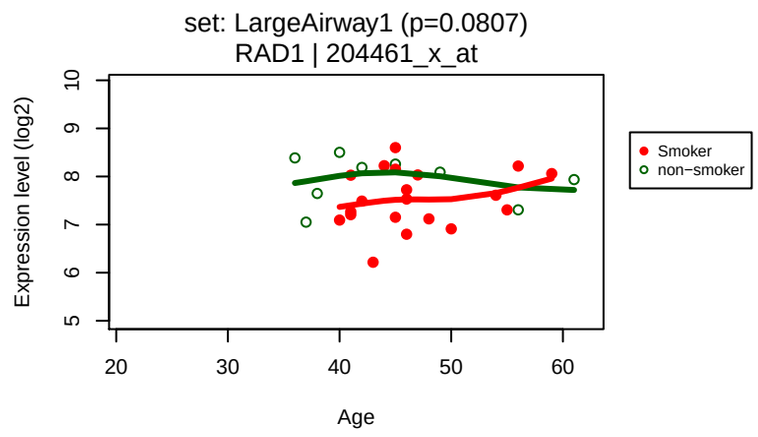

Gene expression of smokers and nonsmokers
gene: RAD1 | 204461_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1718
p=0.2718
p=0.9935
p=0.0807
p=0.4169
p=0.0316
p=0.0651
p=0.3328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

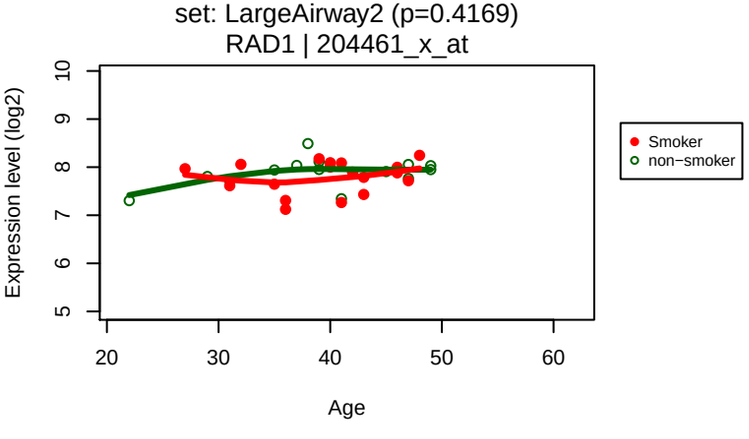

Gene expression ratio (smokers/nonsmokers)
gene: RAD1 | 204461_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1718
p=0.2718
p=0.9935
p=0.0807
p=0.4169
p=0.0316
p=0.0651
p=0.3328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

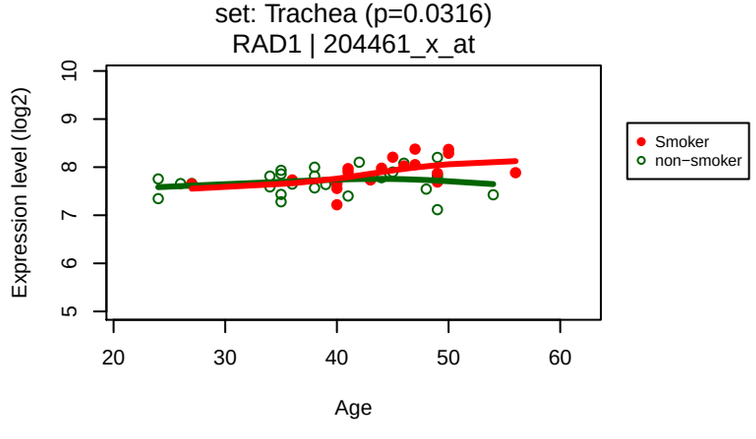

Gene expression ratio in Males
gene: RAD1 | 204461_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3594
p=0.038
p=0.1166
p=0.335
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD1 | 204461_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7714
p=0.0944
p=0.6028
p=0.86
SmallAirway2
LargeAirway2
Trachea
Macrophage2

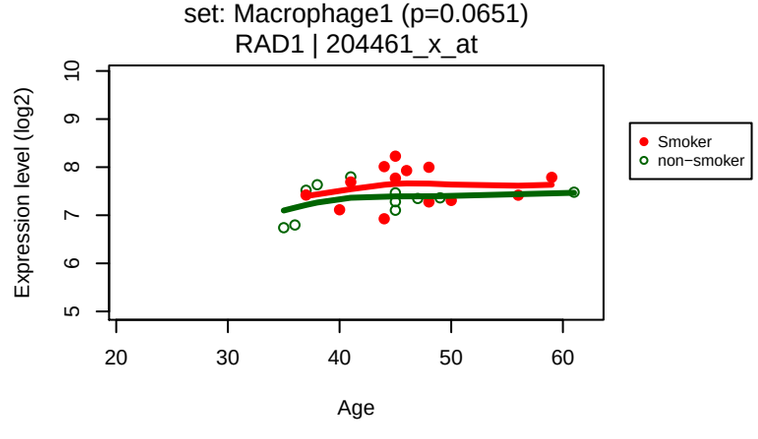

set: SmallAirway1 (p=0.1718)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker

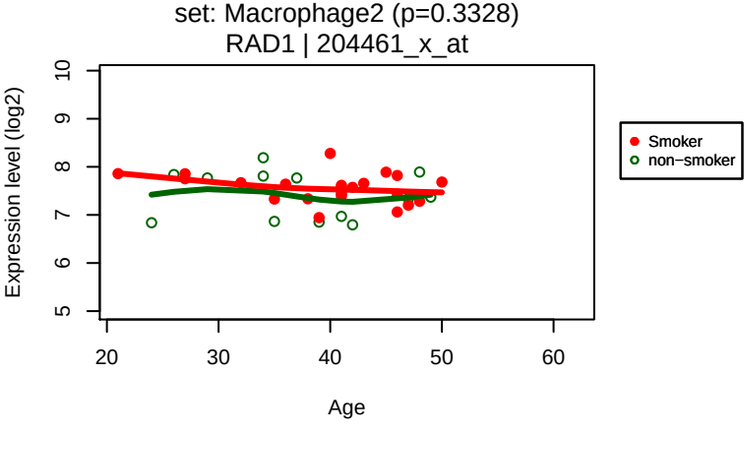

set: SmallAirway2 (p=0.2718)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker

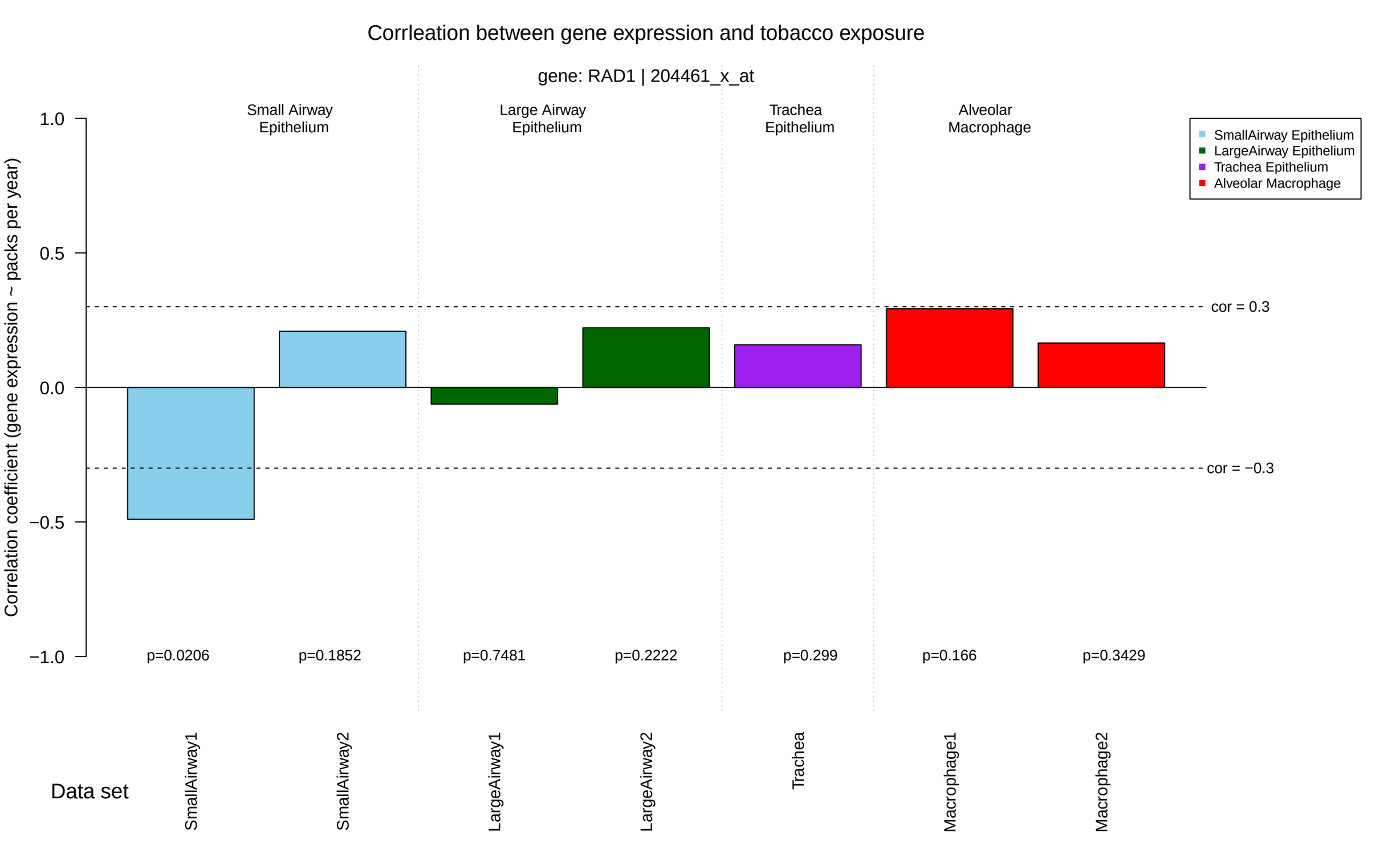

set: LargeAirway1 (p=0.0807)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4169)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0316)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0651)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3328)
RAD1 | 204461_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD1 | 204461_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0206
p=0.1852
p=0.7481
p=0.2222
p=0.299
p=0.166
p=0.3429
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2